BUSINESS INTELLIGENCE - FINAL EXAM

End-to-End Machine Learning Pipeline

Professors: Ph.D. Muñoz-Herrera & M. Sc. Hormazábal-Vildósola

Section	Professor	Course code	UDD e-mail
1	Ph.D. Muñoz-Herrera	(IIB415A-1) [2026-1-1]	semunoz@udd.cl
2	M. Sc. Hormazábal-Vildósola	(IIB415A-2) [2026-1-1]	hector.hormazabal@udd.cl
Release	June 22, 2026	**Due Date**	July 10, 2026

GUIDELINES

1. Evaluation Structure & Grading

This case study is the final examination for the course and contributes **30% of your final grade**. The remaining 70% corresponds to your presentation grade. Mastery of this assessment requires demonstrating both technical proficiency and professional communication skills.

2. Core Objective

The primary goal is to evaluate your ability to autonomously manage an end-to-end Machine Learning pipeline. You are expected to interpret a complex business case and translate it into a functional analytical solution, ensuring that every decision is grounded in business context and validated by established performance metrics.

3. Technical Pipeline Requirements

You must execute a rigorous workflow, covering the following mandatory stages:

- **Data Ingestion & Integrity:** Efficient reading of the dataset, performing thorough "Data Sanity Checks" to identify anomalies, noise, or logical errors.
- **Sanitization:** Systematic correction of inconsistencies (missing data, duplicates, outliers, and physical impossibilities, etc).
- **Data Pre-processing:** Execution of feature engineering, including scaling, categorical encoding, and necessary transformations.
- **Modeling & Evaluation:** Training and evaluation using the various Machine Learning tools covered during the semester.
- **Business-Driven Conclusion:** Your findings must be actionable, linking performance metrics directly to business objectives.

4. Group Registration

Groups must consist of exactly two students. **You must register your team by June 27th by sending an email with the full names and email addresses of both team members.** For students in Section 1, please register via email to: `semunoz@udd.cl`. For students in Section 2, please register via email to: `hector.hormazabal@udd.cl`. **Only registered groups will be considered for the final evaluation; no late registrations will be accepted**.

5. Submission Requirements

Only one member per group is authorized to upload the following deliverables via CANVAS platform:

- **Jupyter Notebook:** A single `.ipynb` file named `BI_Final_Exam_Apellido1_Apellido2_SX.ipynb`.
  Naming Convention: Surnames must be sorted alphabetically. X represents your section number.
- **Video Presentation:** A video of **maximum 15 minutes** (format: `.mp4` or equivalent). The notebook must remain visible on screen throughout the recording. You must detail your workflow, results, analysis, and recommendations in **English (mandatory)**.

1. Analytical framework & Business context

The following section outlines the critical business challenges that LogiFast Latam is currently facing. As an ML Consultant, you are expected to analyze the provided operational context to identify key inefficiencies and strategic opportunities. This document serves as the foundational brief for the development of your predictive transformation pipeline, ensuring that all subsequent analytical efforts remain strictly aligned with the firm's core business objectives and performance requirements.

BUSINESS CASE: LOGIFAST LATAM

LogiFast Latam stands as a dominant force in the regional last-mile logistics ecosystem, operating across highly fragmented urban and rural geographies. Handling over 10,000 shipments per year, the company has scaled aggressively by leveraging a high-density network of distribution centers. However, the current landscape is increasingly hostile: regional competitors are aggressively undercutting prices, and global logistics tech-players are entering the market with advanced automated dispatch platforms. LogiFast finds itself at a crossroads; while the scale of operations is significant, the company's internal intelligence remains tethered to legacy, reactive dashboards that track historical performance without providing foresight into future market volatility. This structural lag in decision-making—exacerbated by thin operational margins and the high cost of customer acquisition—has led to a stagnation in growth, an increase in churn among high-value contract clients, and significant inefficiencies in resource allocation across a fleet that frequently operates under capacity.

As the appointed ML Consultant, you are tasked with leading the "LogiFast Predictive Transformation Project". The goal is to integrate Artificial Intelligence, specifically leveraging Machine Learning, as its core predictive engine, but also as the foundational architecture for strategic planning. LogiFast possesses a massive, high-volume repository of historical shipping data, operational performance logs, and business client metadata. Your objective is to build a robust analytical pipeline that cleanses, standardizes, and leverages this data to transition the firm from "reporting-based" management to ML-driven approach. This is not about generative automation, but about scientific, model-based decision-making. This transformation must directly address three core strategic fronts:

- **Client Segmentation and Market Understanding:** The company currently struggles with a "one-size-fits-all" commercial approach. You must uncover the latent behavioral structures within the client base to define distinct, actionable archetypes. By transitioning from simple demographic categorization to complex behavioral segmentation, the firm will be able to tailor its service models and loyalty incentives, ensuring that the most valuable and stable business accounts receive prioritized attention, thereby defending market share against emerging competitors.
- **Service Continuity and Risk Management:** Client attrition represents the single greatest threat to the firm's long-term sustainability. The business requires a proactive "early-warning" system that identifies shifts in service reliability long before they manifest as formal service cancellations or contract terminations. The challenge is to identify, through operational performance patterns, the critical failure points that drive clients away, enabling the operations department to intervene with targeted service recovery programs.
- **Financial Optimization and Capacity Planning:** In an era of volatile fuel prices and fluctuating demand, the firm must move from static budget planning to dynamic, value-based forecasting. By projecting the economic potential and future operational costs of each client account, the finance team aims to dynamically adjust credit risk exposure and optimize fleet capacity. This will ensure that operational assets—drivers, vehicles, and routing schedules—are utilized with maximum efficiency, aligning every delivery with the firm's bottom-line profitability and long-term valuation goals.

To achieve these goals, you will address these business challenges by designing a comprehensive machine learning pipeline, acting as the bridge between raw data and executive strategy. You will utilize advanced statistical and predictive techniques to extract actionable patterns from the data, with the objective of providing data-backed recommendations that are both technically rigorous and commercially sound. These insights will empower the leadership team to implement strategic interventions that optimize resource allocation, enhance client retention, and drive measurable improvements in the company's bottom-line profitability, ensuring the long-term resilience of LogiFast Latam in a competitive and demanding regional market.

2. Dataset specifications: LOGIFAST OPERATIONS

The LogiFast Latam operational dataset encompasses transactional and behavioral logs captured across all regional distribution hubs in Chile, Peru, and Colombia. The data spans from **January 1, 2025, to March 31, 2026**, representing a period of significant scaling and logistical restructuring. The logs record every shipment lifecycle event, client interaction, and fleet telemetry point, resulting in a high-density repository of over 10,000 records across 20 distinct variables.

Column	Var	Type	Subtype	Unit	Description	Example
`client_id`	-	Qualitative	Nominal	-	Unique business client identifier	CLI-8829
`region`	X_1	Qualitative	Nominal	-	Operational geographic zone	North
`shipping_service`	X_2	Qualitative	Nominal	-	Service tier of the shipment	Express
`shipment_priority`	X_3	Qualitative	Ordinal	-	Priority scale (1 to 5)	4
`vehicle_type`	X_4	Qualitative	Nominal	-	Primary fleet asset used	Truck
`total_ship_vol`	X_5	Quantitative	Discrete	Units	Parcels shipped per month	1240
`avg_parcel_weight`	X_6	Quantitative	Continuous	kg	Avg weight per parcel	15.2
`num_shipments`	X_7	Quantitative	Discrete	Count	Monthly transaction frequency	85
`app_activity`	X_8	Quantitative	Continuous	Hours	Dashboard usage per month	12.4
`years_as_client`	X_9	Quantitative	Continuous	Years	Business relationship tenure	3.5
`reliability_score`	X_{10}	Quantitative	Discrete	Score	Internal on-time index	780
`delivery_dist`	X_{11}	Quantitative	Continuous	km	Avg route distance	45.2
`complaints_filed`	X_{12}	Quantitative	Discrete	Count	Number of service complaints	2
`fuel_consumption`	X_{13}	Quantitative	Continuous	Liters	Fuel efficiency metric	110.8
`weekend_ship_ratio`	X_{14}	Quantitative	Continuous	Ratio	Proportion of weekend orders	0.15
`customer_segment`	X_{15}	Qualitative	Ordinal	-	Small, Medium, Large	Medium
`warehouse_id`	X_{16}	Qualitative	Nominal	-	Origin distribution center ID	WH-001
`delay_risk_flag`	X_{17}	Qualitative	Binary	-	Risk level (High/Low)	High
`avg_delivery_time`	X_{18}	Quantitative	Continuous	Hours	Average delivery time	4.2
`client_retention_flag`	Y_1	Qualitative	Binary	-	**Classification Target:** Binary indicator (1 churned, 0 otherwise)	1
`total_deliv_cost`	Y_2	Quantitative	Continuous	USD	**Regression Target:** Total operational cost per account	4500.75

These logs were extracted directly from the LogiFast Latam ERP and Tracking Systems and are presented here as a structured, comma-separated repository.

Data Integrity & Sanitization Challenge

The provided dataset contains raw operational logs. Before modeling, you must audit the data to ensure its reliability, as your predictive engine's performance depends on the quality of your input. Address the following:

- **Structural Completeness:** Audit missing information; implement a robust imputation or removal strategy.
- **Data Integrity:** Identify and eliminate redundant records to ensure unique event representation.
- **Standardization:** Scrutinize categorical data for label inconsistencies or encoding variations.
- **Logical Validity:** Define boundaries to filter out values that violate physical or operational constraints.
- **Statistical Anomalies:** Analyze extreme variance; justify the treatment of outliers as genuine edge cases or logging errors.

Document your diagnostic assessment and rationale within your notebook.

3. Dataset access

This section provides the necessary entry point to the LogiFast operational environment. You are provided with a dedicated data repository containing historical shipping logs, contract metadata, and performance metrics. It is your responsibility to ensure that the data is loaded correctly and remains consistent throughout the initial inspection. Pay close attention to the output of `df.info()`, as it will be your first indicator of potential data quality issues, such as missing values, incorrect data types, or structural anomalies that must be addressed before proceeding to the cleaning and preprocessing phases.

Python Access: Loading the Dataset

```python
import pandas as pd

# Load the operational dataset - NO NEED TO CHANGE ON YOUR MACHINE
path = "url"
df = pd.read_csv(path)

# Quick data inspection
print(f"Dataset Shape: {df.shape}")
print(df.info())
```

Note on Data Status

The provided dataset has not undergone any cleaning, transformation, or pre-processing stages; it resides in our public GitHub repository as raw, unprocessed operational logs. These logs were extracted directly from the LogiFast Latam ERP and Tracking Systems and are presented here as a structured, comma-separated repository.
As raw data, this dataset contains the inherent inconsistencies, noise, and artifacts typical of high-velocity logistical systems. We have intentionally preserved these imperfections, to ensure that your analysis simulates a professional data engineering and science workflow. It is your responsibility to assess the state of the data, document your findings, and execute the necessary sanitization protocols before integrating these variables into any models.

4. Project: Technical Requirements and Score Distribution

Technical Deliverables: Project Pipeline Roadmap

You must document your work following this mandatory roadmap. These represent the minimum requirements; additional analyses or models are encouraged to enhance your project's performance. The table below outlines the mandatory deliverables and the corresponding score distribution, which totals 50 points. This distribution reflects the technical complexity of each stage within the pipeline, ensuring that efforts are aligned with the critical analytical requirements of the project.

Stage	Mandatory Deliverables	Points
1. Exploration	Descriptive stats, distribution plots, and correlation matrix.	5
2. Data Audit	Identify errors: nulls, duplicates, outliers, and categorical inconsistencies.	5
3. Data Fixing	Sanitization report and the generation of a clean, processed dataset.	5
4. Preparation	Feature engineering: scaling, encoding, transforming and training/testing partitioning.	5
5. Modeling	**Class/Regr:** Decision Tree, Random Forest. **Cluster:** K-Means + PCA.	10
6. Optimization	Hyperparameter tuning (GridSearch/RandomizedSearch).	8
7. Evaluation	**Class:** Acc, Prec, Rec, F1, AUC. **Regr:** MSE, RMSE, MAE, MAPE. **Cluster:** Silhouette.	7
8. Reporting	Final results, model selection rationale, and business recommendations.	5
	Total	**50**

Note: *All stages must be evidenced in your Jupyter Notebook. Clearly justify your methodological decisions and explain the business impact of your findings.*

5. Best Practices and Professional Standards

To ensure the professional quality of your work, the following standards are mandatory throughout the project. These criteria serve as the foundation for your evaluation and are essential for demonstrating technical rigor:

- **Jupyter Notebook Structure:** Organize your notebook using clear Markdown headers for every stage. Each section must be self-contained, with narrative text explaining the purpose of the code blocks and the interpretation of the results.
- **Exploratory Data Analysis (EDA):** Descriptive statistics and plots must include clear titles, axis labels, and legends. Do not simply print raw outputs; explain what the patterns in the data suggest regarding the project's objectives.
- **Performance Measurement:** Methodological choices must be justified by analytical evidence. You are expected to select performance metrics that are appropriate for the specific problem (e.g., justifying why F1 score or RMSE is chosen over other alternatives).
- **Code Quality:** Use clear, descriptive variable names and comment on your code where necessary. The pipeline must be reproducible, and the sanitization report should clearly document how data issues were addressed.
- **Business Impact:** Every analytical finding must be linked back to a business recommendation. The final report is not just a summary of technical steps, but a decision-support tool.

6. Grading Scale

The following table details the conversion from raw points (Total: 50) to the usual grading scale (1.0 to 7.0), calculated with a 60% requirement.

Pts	Grade	Pts	Grade	Pts	Grade	Pts	Grade
0	1.0	13	2.3	26	3.6	39	5.9
1	1.1	14	2.4	27	3.7	40	6.1
2	1.2	15	2.5	28	3.8	41	6.2
3	1.3	16	2.6	29	3.9	42	6.3
4	1.4	17	2.7	30	4.0	43	6.5
5	1.5	18	2.8	31	4.5	44	6.6
6	1.6	19	2.9	32	4.7	45	6.8
7	1.7	20	3.0	33	4.9	46	6.9
8	1.8	21	3.1	34	5.1	47	7.0
9	1.9	22	3.2	35	5.3	48	7.0
10	2.0	23	3.3	36	5.4	49	7.0
11	2.1	24	3.4	37	5.6	50	7.0
12	2.2	25	3.5	38	5.7		

Good luck!